

Rohit Suryaa Saravanan

Software Developer & Data Scientist

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Available from 15 July 2026 · Open to relocating



SUMMARY

MSc Data Science student with two peer-reviewed IEEE publications in applied machine learning, 1+ year of professional software experience, and hands-on Agentic AI delivery through a LangGraph/RAG luxury-vehicle concierge built with local LLM tooling. Track record spans data exploration, prototyping, model evaluation, stakeholder-facing dashboards, and production-minded software delivery. English fluent (C1+); German A2 and actively improving.

EXPERIENCE

Associate Software Developer — EigenSecure (Remote, USA)

Mar 2024 – Apr 2025

- Translated business and security requirements into backend features, RESTful APIs, and proof-of-concept components within an Agile/Scrum cycle, collaborating with cross-functional stakeholders across MySQL, PostgreSQL, and MongoDB data environments.
- Optimised SQL queries, data workflows, and CI/CD pipelines to support reliable production deployments and traceable audit trails.
- Communicated implementation status and trade-offs to both engineering and product stakeholders, balancing security constraints with delivery cadence.

SELECTED PROJECTS

Luxury Concierge Agent — Agentic AI for bespoke vehicle configuration

2026

- Built a local-first agentic AI concierge that translates client usage, region, cabin mood, budget, and delivery timeline into guided luxury-vehicle configuration recommendations.
- Implemented a LangGraph multi-step agent with local RAG over handcrafted product knowledge, mocked configuration/pricing/availability tools, conversation memory, a Typer eval CLI, pytest checks, and a polished Streamlit UI.
- Stack:** Python, LangGraph, Streamlit, Ollama/qwen3.5:9b, RAG, ChromaDB-ready retrieval, Typer, pytest · github.com/srohitsuryaa21/luxury-concierge-agent

Supply Chain Intelligence — End-to-End AI Pipeline

2025

- Delivered an end-to-end analytics pipeline across 110,123 e-commerce transactions: from data exploration and feature engineering, through SARIMA forecasting at sub-20% MAPE and EOQ-based inventory optimisation, to delivery analytics across 92.1% on-time orders and seller-risk scoring for 881 sellers.
- Surfaced findings to operational stakeholders through an interactive Plotly Dash decision dashboard.
- Stack:** Python, Pandas, NumPy, SARIMA, XGBoost, Plotly Dash, SQL, statistical analysis · github.com/srohitsuryaa21/supply-chain-forecasting

Windmill Pitch Prediction with Adaptive Control — IEEE ICEMCE 2023

Feb – Jun 2024

- Reduced modelled forecasting error from 8.6% to 2.3% and increased modelled energy output by 21% by combining XGBoost prediction with a residual-correction layer over six months of turbine operational data.
- Engineered 26 temporal features and validated the model via simulation and statistical evaluation, achieving 96% prediction accuracy under varying operational conditions.
- Stack:** Python, XGBoost, time-series feature engineering, simulation, statistical evaluation · [DOI: 10.1109/ICEMCE57940.2023.10433983](https://doi.org/10.1109/ICEMCE57940.2023.10433983)

DeFi Farmer Platform — IEEE ICSCSS 2024

2024

- Designed a blockchain-based transparent agricultural finance system using smart contracts to automate loans, payments, insurance, and supply-chain traceability for smallholder farmers.
- Built Ethereum contracts with a Web3.py backend and local Ganache validation; modelled digital-transparency adoption rising from below 50% to about 80%.
- Stack:** Ethereum, Solidity, Web3.py, Ganache, smart contracts · [DOI: 10.1109/ICSCSS60660.2024.10625462](https://doi.org/10.1109/ICSCSS60660.2024.10625462)

Employee Attrition Prediction — Explainable ML for HR

Dec 2022 – Feb 2023

- Built a classification pipeline on a public HR dataset: EDA, feature engineering, model comparison, and explainable driver analysis, enabling HR stakeholders to act on retention risk.
- Stack:** Python, Scikit-learn, Pandas, EDA, classification, explainability · github.com/srohitsuryaa21/employee-attrition-prediction

Personal Portfolio — rohitsuryaa.com

2025 – Present

- Designed and deployed a bilingual EN/DE production web portfolio with an ASP.NET Core Minimal API backend, EF Core, and PostgreSQL via Npgsql, with automated CI/CD to GitHub Pages through a GitHub Actions workflow.
- **Stack:** C#, Blazor WebAssembly, ASP.NET Core Minimal API, EF Core, PostgreSQL, GitHub Actions · rohitsuryaa.com

EDUCATION

MSc Data Science — Hochschule Fulda, Germany

Apr 2025 – Present

Focus: Machine Learning, Big Data & Distributed Systems, Data Mining & Predictive Analytics, Process Mining, Data Visualisation & BI, Research Methods, Data Ethics & GDPR.

BTech Big Data Computer Science — SRM University, Chennai, India

Aug 2020 – Nov 2024

Focus: Big Data systems, Distributed Systems, Data Structures & Algorithms, Database Management, Software Engineering.

TECHNICAL SKILLS

Python & ML: Python, Pandas, NumPy, Scikit-learn, XGBoost, PyTorch, TensorFlow, SARIMA, time-series, classification, explainability, statistical analysis

AI Delivery: Agentic AI workflows, LangGraph agents, Retrieval-Augmented Generation, local LLM tooling with Ollama, eval harnesses, Streamlit demos, stakeholder communication

Visualisation & BI: Plotly Dash, Power BI, Tableau, dashboard design for non-technical audiences

Data Engineering: ETL pipelines, complex SQL, data preprocessing, distributed-systems foundations

Cloud & DevOps: Azure, Docker, GitHub Actions, Jenkins, CI/CD, Git, YAML, Linux

Databases: PostgreSQL, MySQL, MongoDB

Software Engineering: RESTful APIs, ASP.NET Core, EF Core, Blazor, Spring Boot, Flask, JWT, Git workflows

LLM Tooling: LangGraph, RAG, Ollama/qwen3.5:9b, ChromaDB-ready retrieval, Typer eval CLI, pytest

CERTIFICATIONS

IBM Data Science Professional (Coursera) · Airbus Supply Chain Specialist

LANGUAGES

English: Fluent (C1+) · German: A2 (actively improving) · Tamil: Native